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Chapter 1

INTRODUCTION

1.1 Overview

Modern warfare has transitioned into an increasingly data-driven environment where split-second decisions are critical for survival and mission success. The sheer volume of visual information from surveillance feeds, drones, and tactical environments often overwhelms personnel, leading to cognitive fatigue and a higher margin for error. Manual threat assessment is no longer sufficient to ensure the safety and tactical superiority of ground units in high-stress scenarios. Consequently, there is an urgent need for intelligent, autonomous systems that can process vast amounts of real-time visual data to provide actionable insights and enhance situational awareness.

VeerDrishti is an AI-enabled combat vision system designed to bridge this gap by providing soldiers with a persistent, intelligent observer through a camera-integrated solution. By leveraging state-of-the-art computer vision models and edge computing, the system autonomously identifies enemy combatants, vehicles, and weapons. The primary objective is to offload the mental burden of constant monitoring from the soldier, allowing them to focus on strategic execution. Developed as a lightweight solution suitable for deployment on wearable hardware, the system aims to transform how visual data is utilized on the front lines of national security.

The project incorporates sophisticated methodologies to ensure reliability under diverse and challenging battlefield conditions. It utilizes multimodal image fusion—combining visible and thermal spectrums—to maintain high detection accuracy in low-light or obscured environments. Furthermore, by implementing edge-optimized architectures and advanced feature fusion techniques, the system achieves high-speed inference without compromising precision. This integration of versatile object detection and robust information processing ensures that the system remains a reliable asset, offering a significant technological advantage in modern combat and defense operations.

1.2 Problem Statement

The problem statement is to address the challenges of managing vast real-time visual data in modern battlefield environments, where manual monitoring is slow, error-prone, and inefficient; in order to enhance situational awareness and tactical decision-making, an AI-powered combat vision system is proposed to automatically detect, classify, and assess threats in real time, improving response speed and operational efficiency.

1.3 Motivation

The motivation for VeerDrishti is to minimize casualties and save lives by providing an intelligent, persistent observer in combat environments. Modern warfare involves overwhelming visual data that leads to cognitive fatigue, increasing the risk of fatal human error. The system serves as a technological safeguard, enhancing personnel safety and protecting national security forces during high-stress operations.

1.3.1 Scope

The scope of the project is to develop an autonomous, real-time vision system designed to detect enemy personnel, weapons, and vehicles in tactical environments. The technical implementation will focus on developing a lightweight system, optimized for localized inference on mobile edge devices.

1.3.2 Purpose

The purpose of the project is to provide an autonomous, intelligent observer that delivers real-time situational awareness in high-stress combat environments. By automating threat detection and mitigating cognitive fatigue, the system serves as a technological safeguard dedicated to achieving the project's core mission: minimizing casualties and saving lives.

1.3.3 Applicability

The project will be applicable in active combat operations, border surveillance, and high-risk tactical missions where ground units require instantaneous threat identification. The edge-optimized architecture will ensure reliability in signal-denied zones and low-visibility environments, serving as a critical force multiplier for national security personnel.

Beyond defense, the system's ability to automate hazard detection will be directly transferable to emergency response and disaster management, fulfilling the mandate to minimize casualties and save lives.

1.4 Objectives

The project's primary objectives are to develop a lightweight, real-time combat vision system that can be deployed on wearable hardware, capable of autonomously detecting and classifying threats in dynamic battlefield environments.

Specific objectives for the project include:

i. Implement a lightweight real-time object detection model.

ii. Enable on-screen visual highlighting of identified threats.

iii. Optimize system performance for low latency and smooth operation.

iv. Evaluate accuracy and response time under simulated field conditions.

1.5 Methodology

1.6 Sustainable Development Goal Relevance

The project aligns with United Nations Sustainable Development Goal 16: Peace, Justice, and Strong Institutions.

1.6.1 Target 16.1: Significantly reduce all forms of violence and related death rates everywhere

VeerDrishti will enhance battlefield situational awareness by detecting and tracking potential threats in real time. This capability will enable defense personnel to respond faster and make more informed decisions during combat situations. By improving awareness and reaction time, the system will help reduce operational risks and potentially decrease casualties.

1.6.2 Target 16.3: Promote the rule of law at the national and international levels and ensure equal access to justice for all

VeerDrishti will record visual data during operations, enabling improved mission analysis and operational review. It will support evidence-based evaluation of decisions made during missions. Such recorded data could also assist investigations related to violations of international humanitarian law, including potential war crimes.

1.6.3 Target 16.A: Strengthen relevant national institutions, including through international cooperation, for building capacity at all levels, in particular in developing countries, to prevent violence and combat terrorism and crime

VeerDrishti will integrate AI-based detection and tracking technologies, strengthening the technological capacity of national security institutions. The project will support peacekeeping operations and the protection of national interests.

1.7 Feasibility Study

The feasibility of the proposed combat vision system is assessed across technical, operational, and economic dimensions.

1.7.1 Technical Feasibility

The proposed system is technically feasible using existing smartphone hardware and computer vision frameworks. Modern phones provide high-resolution cameras, sufficient processing power, and GPU support for real-time object detection. Lightweight AI models can be optimized for edge deployment, enabling on-device threat detection without requiring complex external hardware.

1.7.2 Operational Feasibility

The system can be integrated into existing helmet setups with minimal modification, using a mounted smartphone as both camera and display interface. It reduces cognitive load by automating threat detection and visual highlighting. Basic training would enable personnel to interpret AR overlays effectively in dynamic environments.

1.7.3 Economic Feasibility

Using commercially available smartphones significantly lowers development and deployment costs compared to specialized military hardware. Open-source AI frameworks further reduce software expenses. The prototype approach minimizes initial investment, making the system cost-effective for research, testing, and potential scalable improvements in future versions.

1.8 Work Distribution

Table 1.1: Work Distribution Among Team Members

Name	Tasks
Gaurav Chauhan	Literature Survey and Report Writing
Gaurav Singh	Literature Survey and Slide Preparation

Chapter 2

LITERATURE REVIEW

2.1 Base Paper

2.1.1 Military target detection method based on EfficientDet and Generative Adversarial Network

Authors: X. Zhuang, D. Li, Y. Wang, and K. Li
Journal: Engineering Applications of Artificial Intelligence
Published: June 2024
DOI: 10.1016/j.engappai.2024.107896

Summary: This paper proposes a novel method for military target detection by integrating EfficientDet, a state-of-the-art object detection model, with a Generative Adversarial Network (GAN) to enhance the training data. The authors demonstrate that this approach significantly improves detection accuracy, especially in scenarios with limited training data. The method is evaluated on a military target dataset, showing superior performance compared to traditional detection methods.

Methodology: The authors first use EfficientDet to detect targets in the original dataset. Then, they employ a GAN to generate synthetic images of military targets, which are added to the training set. The enhanced dataset is used to retrain the EfficientDet model, resulting in improved detection performance.

Results: The method achieved 92.8% mean Average Precision (mAP) on a custom military dataset, outperforming YOLOv3 and YOLOv5. It maintained real-time performance (15 FPS) on the test hardware. In complex scenarios (dim lighting, occlusions), the model still achieved over 90% detection accuracy.

2.2 Supplementary Papers

2.2.1 Real-Time Versatile Object Detection With High Accuracy And Reduced Information Loss During Wartime

Authors: D N, Shylaja and Dhanalakshmi, M.
Conference: 2024 International Conference on Integration of Emerging Technologies for the Digital World (ICIETDW)
Published: September 2024
DOI: 10.1109/ICIETDW61607.2024.10939768

Summary: A state-of-the-art object detection system for analyzing camera or drone footage. Designed to identify battlefield threats such as vehicles, weapons, and combatants. Uses a vision-language model, combining images and natural language descriptions for better understanding and classification.

Methodology: CLIP links images with text descriptions, enabling detection of new objects without large labeled datasets. SPPELAN Pooling module that ensures fixed feature representation with minimal information loss. N-gram & GLIP Labeler extracts noun phrases and creates pseudo-boundaries for objects to generate training data. K-means Clustering groups similar image regions to improve efficiency in cluttered scenes.

Results: Compared with Grounding DINO on the Microsoft COCO dataset, the system showed better efficiency. Lower complexity (48M parameters vs 172M), higher accuracy (AP 35 vs 25.6) and faster framerate (17.6 FPS vs 1.5 FPS), making it suitable for realtime deployment.

2.2.2 EDNet: Edge-Optimized Small Target Detection in UAV Imagery - Faster Context Attention, Better Feature Fusion, and Hardware Acceleration

Authors: Song, Zhifan and Zhang, Yuan and Ebayyeh, Abd Al Rahman M. Abu
Conference: 2024 IEEE Smart World Congress (SWC)
Published: December 2024
DOI: 10.1109/SWC62898.2024.00141

Summary: EDNet is a deep-learning framework for detecting small objects in UAV/drone imagery. Built on YOLOv10 and optimized for edge computing, allowing efficient use on mobile and low-power devices. The model is scalable, with 7 sizes ranging from Tiny (lightweight) to XL (high performance).

Methodology: C2f-FCA Block uses faster context attention to extract features with lower computational complexity. WIoUv3 Loss Function helps the model focus on high-quality samples and ignore noisy data. Optimized Deployment uses CoreML, TensorRT, and OpenVINO for high-speed performance across different processors.

Results: EDNet outperformed models like YOLOv8 and YOLOv10 in accuracy and speed. Higher Accuracy: Up to 5.6% improvement in mAP_{50} compared to YOLOv10 with fewer parameters. Better Efficiency: EDNet-M surpassed YOLOv10-X while using 39.6% fewer parameters. Real-Time Performance: 16–55 FPS on iPhone 12; EDNet-Tiny runs 3 times faster on Raspberry Pi 5.

2.2.3 Multi-modal action recognition via advanced image fusion techniques for cyber-physical systems

Authors: Shou, Zaiyong and Zhu, Daoyu
Journal: Frontiers in Physics
Published: August 2025
DOI: 10.3389/fphy.2025.1576591

Summary: A framework for cyber-physical systems such as autonomous vehicles and smart surveillance. It introduces MSAF-Net, a system for human action recognition. It combines multiple visual data types like RGB, depth, and thermal images for more accurate detection.

Methodology: Multi-Scale Attention Fusion extracts features at different detail levels and prioritizes the most relevant ones. Cross-Level Feature Interaction allows network layers to share information, preserving fine details and global structure. Adaptive Fusion Strategy ensures clear and consistent fused data using semantic and structural guidance.

Results: MSAF-Net outperformed existing state-of-the-art models across four major datasets. It achieved 91.54% accuracy on UCF101 and 92.14% on ActivityNet. The model remains robust even if one sensor fails and can be pruned for faster performance on low-power hardware.

2.2.4 Static Early Fusion Techniques for Visible and Thermal Images to Enhance Convolutional Neural Network Detection: A Performance Analysis

Authors: Heredia-Aguado, Enrique and Cabrera, Juan José and Jiménez, Luis Miguel and Valiente, David and Gil, Arturo
Journal: Remote Sensing
Published: March 2025
DOI: 10.3390/rs17061060

Summary: The paper discusses fusion of RGB images and infrared data. It compares early and middle fusion techniques to combine both sources to create a more robust real-time detection system for mobile robots and UAVs in search-and-rescue or security missions.

Methodology: Early Fusion Methods like RGBT, HSVT, VTHS, and VT were tested to combine RGB and thermal data. Alignment correction was implemented via optical-flow-based affine transformation to fix RGB–thermal misalignment in the KAIST dataset. Images weere enhanced by applying CLAHE to improve contrast and grayscale range in thermal and visible images.

Results: Best results came from using separate models for day and night. In night conditions, VT and VTHS fusion performed best, relying on brightness and thermal data instead of color. In daylight, RGBT fusion worked best by using color information effectively. Simple early fusion methods often outperformed complex middle fusion models, which were more prone to overfitting.

2.2.5 Infrared-Visible Image Fusion Meets Object Detection: Towards Unified Optimization for Multimodal Perception

Authors: Xiang, Xiantai and Zhou, Guangyao and Niu, Ben and Pan, Zongxu and Huang, Lijia and Li, Wenshuai and Wen, Zixiao and Qi, Jiamin and Gao, Wanxin
Journal: Remote Sensing
Published: November 2025
DOI: 10.3390/rs17213637

Summary: UniFusOD is an end-to-end framework for infrared–visible image fusion and object detection. It combines visible images (rich texture and color) with infrared images (strong edges in low light). The system produces a single fused image optimized for clear visualization and accurate detection.

Methodology: FRA Mechanism uses multiple convolution kernels to capture both fine details and larger semantic features. UnityGrad Optimization balances fusion and detection gradients, preventing conflicts so both tasks train effectively.

Results: UniFusOD outperformed several state-of-the-art models on multiple datasets. Better Detection: Improved mAP by 0.8 (DroneVehicle) and 1.9 (M3FD). Superior Fusion: Produced images with higher information content and better structure. Robust Performance: Particularly effective at detecting small targets in complex environments.

2.3 Comparison Table

Table 2.1: Work Distribution Among Team Members

Sl. No.	Paper Title	Objective	Methodology	Core Idea
[1]	Military target detection method based on EfficientDet and Generative Adversarial Network	To detect targets in low visibility, noisy, or data-scarce combat zones	EfficientDet-D0 + Dual-Discriminator GAN	Uses GANs to generate synthetic military training data and BiFPN for feature fusion
[2]	Real-Time Versatile Object Detection With High Accuracy and Reduced Information Loss During Wartime	To detect unseen or new battlefield threats without manual re-labeling	CLIP Encoder + Modified C2f-Darknet	Uses pseudo-boundary labeling for zero-shot detection and combines vision and language
[3]	EDNet: Edge-Optimized Small Target Detection in UAV Imagery — Faster Context Attention, Better Feature Fusion, and Hardware Acceleration	To enable high-speed detection of tiny objects on low-power drone hardware	YOLOv10 + Faster Context Attention (C2fFCA)	Uses XSmall detection layer for small object detection with hardware acceleration optimizations

Continued on next page

Sl. No.	Paper Title	Objective	Methodology	Core Idea
[4]	Multi-modal action recognition via advanced image fusion techniques for cyber-physical systems	To recognize human/combatant actions accurately even if one sensor fails	MSAF-Net (Multi-Scale Attention-Guided Fusion)	Dynamically prioritizes sensor data and switches to backup sensors when needed
[5]	Static early fusion techniques for visible and thermal images to enhance convolutional neural network detection: A performance analysis	To find the most efficient way to merge RGB and infrared data for robots	YOLOv8 + Early Fusion	Compresses 4-channel data into 3-channel for efficiency processing and uses CLAHE for contrast enhancement
[6]	Infrared-visible image fusion meets object detection: Towards unified optimization for multimodal perception	To solve gradient conflicts when training fusion and detection together	UniFusOD	Use Fine-Grained Region Attention (FRA) mechanism and unified framework to fuse RGB and infrared images for accurate detection

Chapter 3

REQUIREMENTS AND ANALYSIS

The requirements analysis for the VeerDrishti combat vision system encompasses a comprehensive evaluation of functional and non-functional requirements, ensuring that the system meets the operational needs of modern warfare while adhering to technical constraints. The analysis is structured to identify key features, performance metrics, and user interactions necessary for the successful deployment of the system in dynamic battlefield environments.

3.1 Functional Requirements

The functional requirements define the core capabilities of the project, including:

i. **Real-Time Object Detection:** The system must be capable of detecting and classifying combatants, vehicles, and weapons in real time using the integrated camera feed.

ii. **Visual Highlighting:** Detected threats should be visually highlighted on the display interface, providing clear and immediate feedback to the user.

iii. **Open Vocabulary Recognition:** The system should be able to identify novel or previously unseen threats using natural language supervision and language-vision encoders.

iv. **User Interface:** The system must provide an intuitive user interface that allows easy interpretation of the visual information to make informed decisions without distraction.

v. **Data Logging:** The system must provide secure local storage of detection events and video metadata for post-mission analysis and system auditing.

3.2 Non-Functional Requirements

The non-functional requirements address the performance, reliability, and usability aspects of the project, including:

i. **Low Latency:** The system must operate with minimal latency to ensure timely threat detection and response in high-stress combat scenarios.

ii. **Robustness:** The system should maintain high detection accuracy under varying environmental conditions, such as low light, occlusion, and dynamic backgrounds.

iii. **Scalability:** The system should be designed to accommodate future enhancements, such as additional threat categories or integration with other battlefield systems.

iv. **Security:** The system must ensure the confidentiality and integrity of the data collected, preventing unauthorized access and ensuring compliance with military data handling standards.

v. **Usability:** The system should be user-friendly, requiring minimal training for effective operation, and should not contribute to cognitive overload for the user.

3.3 Hardware Requirements

i. **Development Workstation:** High-performance laptop equipped with a dedicated GPU for code development, dataset preprocessing, and initial model debugging.

ii. **Edge Deployment Device:** High-end smartphone with an integrated Neural Processing Unit (NPU) to host the prototype application and execute real-time local inference.

iii. **Cloud Training Infrastructure:** Scalable cloud-based GPU instances utilized for training complex deep learning architectures and performing large-scale hyperparameter optimization.

3.4 Software Requirements

i. **Operating System:** Linux distribution like Ubuntu 26.04 LTS with stable CUDA/ROS support.

ii. **Programming Languages:** Python 3.9+ for model development, Rust for high-performance edge execution and Kotlin for mobile application development.

iii. **IDE/Tools:** VS Code, Jupyter Notebooks for experimentation, and Git for version control.

iv. **Core Framework:** PyTorch 2.0+ or TensorFlow 2.x (primarily PyTorch for YOLOv8/v10 implementation).

v. **Cloud Platform:** Google Cloud Vertex AI or AWS EC2 for large-scale model training.

Chapter 4

SYSTEM DESIGN

4.1 System Architecture

4.2 System Workflow

Chapter 5

SCHEDULING

5.1 Gantt Chart

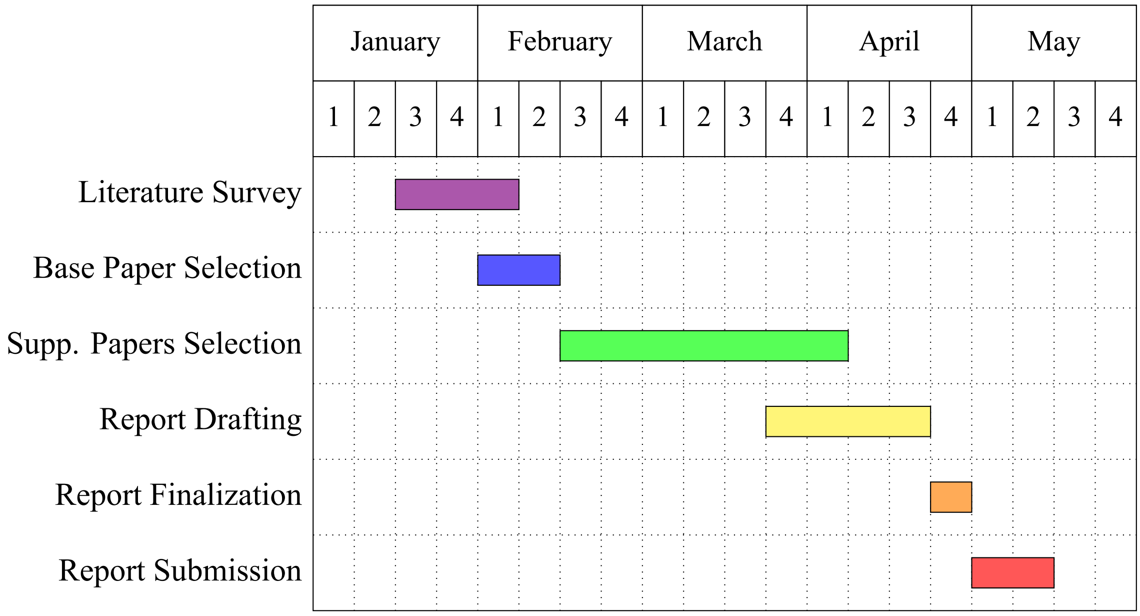

Figure 5.1: Gantt Chart for Major Project (Phase-1)

5.2 Month-wise Progress

5.2.1 January

Conducted a comprehensive literature survey to identify relevant research papers and technologies in the domain of real-time combat vision systems. Identified research papers that could serve as the base paper and foundation for our project.

5.2.2 February

Selected the base paper and conducted extensive reading and analysis to understand the current state of the art, identify gaps in existing solutions, and select a base paper that aligns with our project objectives.

5.2.3 March

Selected 10 supplementary papers that provide additional insights, methodologies, and technologies relevant to our project. These papers contribute to the development of our combat vision system by providing a broader understanding of the field and potential approaches to implementation. Began intial work to draft the project report.

5.2.4 April

Drafted the project report, synthesizing information from the base paper and supplementary papers. Structured the report to clearly articulate the project details and specifications. Iteratively refined the report based on feedback and additional insights gained during the drafting process and drafted the final project report.

5.2.5 May

Submitted the finalized project report, ensuring that all sections are complete, well-organized, and adhere to the required guidelines.

Chapter 6

CONCLUSION

The VeerDrishti project represents a significant advancement in tactical situational awareness by successfully integrating state-of-the-art artificial intelligence with wearable combat technology. By proposing a system that utilizes the lightweight YOLOv8 architecture alongside multimodal image fusion, the project addresses the critical need for real-time, autonomous threat detection in high-stress battlefield environments. The synchronization of visible and thermal spectrums ensures that the system remains operational across diverse and challenging conditions, including total darkness, smoke, and obscured visibility. This comprehensive approach effectively mitigates the cognitive burden on personnel, transforming overwhelming visual data into actionable intelligence and allowing for faster, more accurate strategic decision-making when split-second responses are vital for survival. The technical robustness of the system is further bolstered by the incorporation of sophisticated methodologies such as GAN-based synthetic data augmentation and open-vocabulary detection. By leveraging Generative Adversarial Networks, the project successfully overcomes the limitations of scarce military datasets, while the CLIP-based encoder allows the vision system to adapt to novel, unseen threats through natural language supervision. Furthermore, the emphasis on edge-optimized deployment ensures that high-speed inference can be conducted locally on portable, smartphone-based hardware. This localized processing not only guarantees data privacy and security in signal-denied environments but also ensures that the system provides the low-latency response times necessary for maintaining tactical superiority in active combat scenarios.

Looking forward, the potential for VeerDrishti extends into a scalable framework for future defense innovations. Future iterations could integrate more complex action recognition models to predict hostile intent or incorporate advanced feature-fusion strategies to further enhance the detection of micro-scale targets. In conclusion, VeerDrishti offers a versatile and reliable solution for modern national security. By harmonizing cutting-edge computer vision research with the practical, high-stakes requirements of the frontline, the project provides a definitive technological advantage, ensuring enhanced safety and operational efficiency for ground units in the evolving landscape of modern warfare.

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GLOSSARY